SPEECH-BASED REFLECTIVE SUPPORT MODULE FOR STUDENT STRESS MONITORING

Rajapaksha R.N.D.K.

(IT22315632)

BSc (Hons) degree in Information Technology

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1. Declaration of the candidate & Supervisor

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the non exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

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2. Abstract

Academic stress among university students is dynamic, context-dependent, and strongly influenced by assessment schedules, deadlines, and cumulative workload. Although many digital mental health tools support reflection and coping, most systems provide static assessments that are not transformed into personalized, time-aware predictions. This thesis presents the design, implementation, and evaluation of a focused stress analytics component consisting of three integrated modules: DASS-21 onboarding questionnaire, a personalized stress fingerprint engine, and a stress heatmap linked to academic calendar events for short-horizon stress prediction.

The component establishes a validated psychometric baseline through DASS-21 onboarding. The baseline is then combined with recent check-in patterns and near-term calendar events to produce individualized stress forecasts. Predictions are represented as an interpretable heatmap that highlights low, moderate, and high stress periods over time, enabling proactive self-management. The forecasting pipeline includes data normalization, event-pressure modeling, stress-momentum estimation, variability and recovery features, and feature-attributed fingerprint labeling to identify dominant stress drivers.

This work follows an applied design-and-implementation methodology. The module is developed in a production-oriented mobile and backend architecture and evaluated through functional testing, endpoint verification, metric reporting, and scenario-based stress simulation. In addition to predictive outputs, the component reports confidence and feature-importance values to improve transparency and interpretability. The system supports near-term prediction without requiring invasive personal data, relying instead on onboarding responses, lightweight daily check-ins, and user-maintained academic events.

Keywords *: Stress Fingerprint, Academic Heatmap, Machine Learning, Stress Predictions*

3. Dedication

4. acknowledgement

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Introduction

Stress has emerged as a defining concern for contemporary youth, with university and college students exhibiting some of the highest reported levels. The need to manage simultaneous workloads, interpersonal commitments, and individual adversities keeps the physiological and psychological stress systems activated. Formulating and internalizing sustainable stress-regulatory strategies is, therefore, a university-level imperative: such strategies protect not only the pursuit of Grade Point Average objectives but also the body and mind from cumulative damage, given that heightened alertness is now the backdrop of nearly every educational day [1].

In educational research, students' academic performance and learning pressure are recognized as critical factors influencing both academic success and overall well-being. The concept of *pressure* can be understood in three primary ways: external situations or events that impose demands on individuals, as a state of physical and psychological tension resulting from such demands, and as the physiological and psychological responses triggered by stressful situations.

Within the academic environment, learning pressure refers specifically to the psychological strain experienced by students due to educational demands. These demands include heavy academic workloads, examinations, peer competition, and expectations from parents and teachers. Such pressures have been shown to significantly impact students' mental health, motivation, and academic performance. Prolonged exposure to academic stress can lead to anxiety, burnout, and reduced learning efficiency, highlighting the importance of effective stress management strategies in educational settings [2].

In response to these challenges, this research introduces the MIND+ system, a mobile-based stress management platform designed to support students in managing learning pressure. The system integrates voice journaling and community-based support to help users express emotions, identify stress levels through keyword analysis, and connect with relevant peer groups. By leveraging technology to monitor and address psychological stress, MIND+ aims to enhance students' emotional well-being and improve their academic performance. This approach aligns with modern educational research that emphasizes the integration of mental health support systems within learning environments.

Background Literature

Life stress is a significant risk factor for a wide range of mental and physical health conditions, including anxiety, depression, cardiovascular diseases, and neurodegenerative disorders. Due to its strong influence on overall well-being, accurately measuring stress has become a critical focus in modern research. Traditional methods such as interview-based assessments and self-report questionnaires, although widely used, suffer from limitations including high cost, time consumption, and susceptibility to bias. Biological measures such as cortisol levels and heart rate provide more objective insights but are often invasive and impractical for continuous monitoring.

To address these limitations, recent advancements in digital health have introduced speech-based stress detection as a non-invasive and scalable alternative. Speech carries rich emotional and physiological information, as stress affects vocal characteristics such as pitch, tone, speaking rate, and pauses. These variations occur due to physiological changes in muscle tension and respiration during speech production. By capturing and analyzing these acoustic features, it is possible to infer an individual's stress level in real time [3].

Modern approaches utilize machine learning techniques, particularly deep learning models such as Convolutional Neural Networks (CNNs), to process speech data. Audio signals are first preprocessed using tools like Librosa to extract features such as Mel-Frequency Cepstral Coefficients (MFCCs), chroma features, and spectrograms. These features are then used as inputs to machine learning models that classify emotional states or generate continuous stress scores. The integration of such models into smartphones and smart speakers has made it feasible to monitor stress in natural environments, enabling continuous and real-time assessment [3].

Speech-based stress detection offers several advantages, including low cost, non-intrusiveness, and the ability to capture real-time emotional changes. This makes it highly suitable for applications such as voice journaling systems, where users can express their thoughts verbally while the system automatically analyzes their emotional state. However, despite its potential, this approach faces challenges related to validation, as speech-based stress measures are not yet fully aligned with biological biomarkers. Additionally, environmental noise, individual speech variability, and limited real-world datasets can affect model accuracy [3].

Furthermore, the use of speech data raises significant privacy and ethical concerns. Continuous monitoring through devices such as smartphones and smart speakers introduces risks related to data security, unauthorized access, and misuse of sensitive information. Ethical questions also arise regarding data ownership, consent, and the appropriate use of stress-related insights. Therefore, while speech-based stress detection holds great promise for advancing digital health and voice journaling applications, it is essential to address these technical, ethical, and privacy challenges to ensure responsible and effective implementation.

Previous research [4] has developed the NokoriMe mHealth application to monitor academic stress using a structured questionnaire combined with physiological data such as sleep quality and physical activity. Although the system was found to be intuitive, participants reported that completing lengthy daily questionnaires was time-consuming and could negatively impact user engagement. This highlights a key limitation of manual input-based stress monitoring systems. To address this issue, modern approaches can incorporate voice journaling features, allowing users to express their emotions more naturally and efficiently. Voice-based input reduces user burden while providing richer emotional context, thereby enabling more accurate and continuous stress monitoring.

Previous work [5] proposed a machine learning-based system to classify academic stress levels using standardized questionnaires such as the Perceived Stress Scale (PSS) and Student Academic Stress Scale (SASS). While their approach demonstrated the effectiveness of data-driven stress classification, it still relied heavily on repeated self-reported inputs, which may lead to user fatigue and reduced participation over time. These limitations highlight the need for more user-friendly and continuous data collection methods. To address this issue, modern approaches can incorporate voice journaling features, allowing users to express their emotions more naturally and efficiently. Voice-based input reduces user burden while providing richer emotional context, thereby enabling more accurate and continuous stress monitoring.

Voice journaling systems can be enhanced through speech-based emotion and stress detection, where spoken input is analyzed to identify a user's psychological state. This problem can be framed as a classification task that detects emotional or stress-related patterns from voice data. Early approaches [6] relied on rule-based or lexicon-based techniques, which compare extracted linguistic or transcribed speech content with predefined emotional dictionaries. While these methods are computationally efficient and provide basic indicators of sentiment and stress, they often struggle to capture contextual and nuanced variations in human speech. To improve performance, machine learning approaches have been introduced by combining lexical features with learned representations from data. With the advancement of deep learning, word embedding techniques and sequential models have been widely used to capture semantic and temporal patterns in speech-derived text. However, embedding-based methods alone may not fully represent affective and psychological cues, so combining them with lexicon-inspired emotional features has been shown to improve accuracy and robustness. In the context of voice journaling, this hybrid approach is particularly effective because users express emotions naturally and inconsistently across daily recordings. Overall, integrating lexical features with deep learning-based representations provides a strong foundation for more accurate and context-aware stress and emotion detection in voice journaling applications, supporting mental wellness monitoring and reflective self-analysis.

In conclusion, the reviewed literature highlights that stress and emotion detection from textual and speech data has evolved from simple lexicon-based approaches to more advanced machine learning and deep learning methods. While lexicon-based techniques offer computational efficiency and interpretability, they are limited in capturing contextual and semantic complexity. In contrast, embedding-based models and neural network architectures provide stronger performance by learning richer representations of language. Recent studies further show that hybrid approaches, which combine lexical knowledge with deep learning features, consistently achieve improved accuracy and robustness. However, existing work is often constrained by dataset dependency and limited cross-domain evaluation. Overall, the literature suggests that integrating multiple feature representations and leveraging deep learning techniques offers a promising direction for developing more reliable and generalizable stress detection systems, particularly for applications such as voice journaling and mental well-being monitoring.

Research Gap

Although significant progress has been made in stress and emotion detection from both text and speech data, several important research gaps still exist that limit the effectiveness of current solutions, particularly in real-world applications such as voice journaling systems. Firstly, most existing studies focus on either text-based or speech-based stress detection independently, with limited attention given to integrating these modalities into a unified and practical framework. As a result, current systems often fail to fully utilize the richness of human emotional expression, which is naturally conveyed through both linguistic content and vocal characteristics.

Secondly, many proposed models are developed and evaluated using single-source or domain-specific datasets, such as social media posts, interview transcripts, or controlled experimental data. While these datasets are useful for benchmarking, they do not accurately reflect real-world user behavior or diverse emotional expression across different contexts. This leads to reduced model generalizability when applied to unseen environments, such as mobile voice journaling applications used by students in daily life.

Another major limitation is the heavy reliance on structured or pre-collected data, including questionnaires, labeled datasets, or guided prompts. Although such methods improve model training and evaluation, they impose significant user burden and reduce engagement over time. In contrast, real-world applications require systems that can operate on spontaneous and unstructured user input, particularly in the form of natural speech recordings. Existing systems often fail to address this requirement effectively.

In addition, privacy and ethical concerns remain largely underexplored in current research. Continuous monitoring of speech data introduces risks related to data security, user consent, and sensitive information leakage. However, few studies propose concrete mechanisms for privacy-preserving stress detection systems that can be safely deployed in personal mental health applications.

Finally, there is a lack of comprehensive systems that combine real-time processing, low user effort, emotional intelligence, and cross-context adaptability. Most existing solutions focus on improving classification accuracy rather than building user-centric systems that support continuous emotional tracking and mental well-being support.

Therefore, there is a clear need for a lightweight, privacy-aware, and context-sensitive voice journaling framework that integrates speech-based stress detection with intelligent emotional analysis. Such a system should be capable of handling natural, unstructured speech input while maintaining high accuracy, scalability, and user engagement in real-world scenarios.

Research Problem

Despite significant advances in stress detection from text and speech data, existing approaches still face several limitations that prevent their effective use in real-world, continuous mental health monitoring systems, particularly in applications such as voice journaling for students. Although earlier research has demonstrated the usefulness of lexicon-based methods for identifying stress-related expressions, these techniques largely depend on predefined dictionaries and rule-based structures. As a result, they often fail to capture contextual meaning, subtle emotional cues, and variations in individual language use. This makes them less effective in handling natural, unstructured, and spontaneous user-generated content such as daily voice journal entries.

On the other hand, machine learning and deep learning-based models, including word embeddings and neural networks, have shown improved performance by learning patterns directly from data. However, these models are highly dependent on large, high-quality, and well-annotated datasets. In practice, such datasets are often limited, domain-specific, and not fully representative of real-world scenarios. Many existing studies are also evaluated on single datasets (e.g., social media posts, interviews, or specific corpora), which restricts their generalizability across different user groups, platforms, and contexts. This lack of cross-domain validation raises concerns about how well these models perform when deployed in real-life applications such as student stress monitoring systems.

Furthermore, most existing stress detection systems focus on either textual or speech data independently, rather than integrating both modalities in a unified framework. In the context of voice journaling, speech input contains not only linguistic information but also paralinguistic cues such as tone, pitch, and speaking pace, which are often underutilized when only transcription-based text analysis is performed. Current approaches also struggle to balance accuracy with usability, as many systems require extensive manual input, questionnaires, or structured responses, which reduce user engagement over time.

Another critical issue is the limited adaptability of current models to individual differences. Stress expression varies significantly across users depending on personality, cultural background, and communication style. Most existing models do not adequately account for these variations, leading to reduced performance when applied to unseen users or environments. In addition, privacy and ethical concerns remain a major challenge, especially in continuous monitoring systems that collect sensitive voice data. Ensuring secure data handling, user consent, and responsible usage of stress-related insights is still an open problem in the field.

Therefore, the core research problem addressed in this study is the lack of a robust, scalable, and user-friendly stress detection framework that can effectively analyze voice journaling inputs to identify stress levels in real time. There is a need for a system that combines the strengths of lexical, acoustic, and machine learning-based approaches while remaining generalizable across different datasets and user contexts. Such a system should minimize user effort, improve detection accuracy, and support continuous emotional monitoring without relying heavily on manual questionnaires or domain-specific data constraints.

This research aims to bridge these gaps by developing an integrated approach for stress detection within a voice journaling-based mental wellness system. The proposed solution focuses on improving contextual understanding, enhancing generalization across datasets, and enabling more natural and scalable stress monitoring suitable for real-world student mental health support applications.

Main Objectives

1. To implement a voice journaling and analysis module that allows users to record daily voice entries, detect stress-related keywords, recognize emotional patterns from speech, and calculate corresponding stress levels for continuous mental health monitoring.
2. To design and develop a community support feature that enables users to share posts, express personal experiences, and engage in peer-to-peer support within a safe and interactive environment to reduce stress and improve emotional well-being.
3. To build a secure data management system that stores, retrieves, and tracks user voice recordings, stress scores, emotional history, and usage patterns for long-term analysis and progress monitoring.
4. To ensure fast processing and stable system performance for real-time or near real-time stress detection from voice inputs without affecting user experience.
5. To maintain reliable data handling mechanisms that guarantee consistency, accuracy, and integrity of stored emotional and stress-related information.
6. To provide cross-platform accessibility so that users can access the system seamlessly across multiple devices such as mobile phones, tablets, and web interfaces.
7. To ensure usability and maintainability of the system by designing a user-friendly interface and modular architecture that supports easy updates and enhancements.
8. To implement strong security mechanisms including encrypted data storage and secure access control to protect sensitive user voice data and personal mental health information.